SecEvidence: Litigation Evidence Preservation and Verification on Blockchain with Secure Algorithms

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***Abstract*—The exponential growth of the digital economy has led to electronic commerce, intellectual property rights, and electronic contracts becoming the primary business models for enterprises, which have also resulted in numerous disputes. Using electronic evidence for litigation has become an inevitable trend for courts. However, traditional electronic data faces inherent challenges such as susceptibility to tampering, fragility, and vulnerability to attacks. This undermines the credibility of electronic evidence stored on proprietary or cloud servers, necessitating technical solutions to ensure data integrity. Blockchain technology provides immutable and transparent features, making it a promising solution for ensuring data integrity. This paper proposes a secure litigation electronic evidence preservation and verification framework called `SecEvidence`, leveraging blockchain technology and cryptographic techniques. Our framework designs a robust litigation electronic evidence preservation and verification system, adopts alternative key algorithms to mitigate security risks, and conducts extensive experimental evaluations to validate the feasibility and performance of the system.**

***Index Terms*—Electronic, blockchain, evidence, preservation, verification**

I. Introduction

With the booming development of the digital economy, applications such as e-commerce, intellectual property rights, and electronic contracts are gradually becoming the core business models of enterprises [1]–[3], resulting in an exponential increase in transaction disputes. However, traditional electronic data suffers from inherent issues such as virtuality, fragility, concealment, and susceptibility to tampering. Electronic evidence stored on proprietary or cloud servers is prone to damage during backup and transmission, susceptible to attacks and alterations, resulting in incomplete or compromised evidence, thereby reducing its credibility. Data within the court information system is mostly associated with electronic evidence, thus necessitating technological means to ensure the authenticity and integrity of the data.

Blockchain technology, with its immutable and transparent characteristics, stands as an effective solution for ensuring data integrity [4]. It employs chain data structures for data validation and storage, utilizes distributed node consensus algorithms for data generation and updates, ensures data transmission and access security through cryptography, and establishes a novel distributed infrastructure and computing paradigm via smart contract programming and data manipulation. Within a blockchain system, all participating entities possess identical data chains, preventing unilateral tampering. Modifications to any information require consensus among a predetermined proportion of entities, permitting only the addition of new information while prohibiting the deletion or alteration of existing information. This ensures information sharing among multiple entities and safeguards the immutability and transparency of identities and transactional information.

However, in most consortium blockchains today, the security-critical algorithms are already predetermined, and application developers or users do not have the ability to change these algorithms, leading to the inability of outdated algorithms to meet current security requirements. This may lead to the following issues in electronic evidence in a courtroom scenario: (1) Some encryption algorithms incur high time costs, making it impossible to select appropriate algorithms based on security requirements for the scenario, thus impacting the user experience. (2) In cases where existing algorithms are proven to be insecure, they cannot be updated promptly, posing underlying algorithm security risks. For example, since the collision attack on the sha-1 hash algorithm was cracked in 2017 [5], its security has been compromised, which requires the replacement of other hash algorithms. (3) Using the same algorithm increases the risk of the blockchain system being breached. Adopting different hash algorithms can mitigate this risk, especially concerning the invocation of hash algorithms in signature and HMAC algorithms.

To address this challenge, this paper proposes the secure litigation electronic evidence storage and verification framework, `SecEvidence`. The framework utilizes blockchain technology to compute and record key information of electronic evidence, based on cryptographic techniques such as SM3, SM2P256V1, and SHA512, and employs a multi-node consensus mechanism to maintain electronic evidence, thus achieving technical capabilities such as authenticity, tamper-proof, and traceability of electronic data evidence. Our main contributions are as follows:

- **Designed a litigation electronic evidence storage and verification system.** `SecEvidence` secures electronic materials, business data, and user behavioral information

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in litigation services. It supports verification by handling judges, ensuring the tamper-proof, verifiability, and traceability of electronic data, ensuring the security and credibility of the entire process of litigation service data, and enhancing the authority, professionalism, and judicial credibility of electronic litigation services.

- **Adopt alternative cryptographic algorithms to mitigate potential security risks.** By adopting alternative elliptic curve signatures and hash algorithms at the underlying level of the blockchain system, `SecEvidence` helps avoid potential security risks. It also allows users to adopt alternative or optimized cryptographic algorithms based on varying security requirements, thus reducing the overall time costs of the system.
- **Conducted extensive experimental evaluations to verify the feasibility and performance of the system.** Experimental results demonstrate that the performance of `SecEvidence` is primarily related to the algorithms themselves. Enhancing the security of critical algorithms does not affect the system's operational performance.

II. Related Work

Haridas et al. [6] proposed LegalLedger, a solution to digitize court processes and store them on the blockchain, preventing tampering and providing secure access. NyaYa [7] introduced a blockchain-based electronic evidence management solution, addressing various stages from case registration to hearings and resolution, addressing vulnerabilities such as data loss, disk failures, and trust issues. Ranjan et al. [8], from a different perspective, utilized court case management systems to generate smart contracts that support automated arbitration of legal issues and reduce the cost of dispute arbitration. JusticeChain [9] utilized blockchain to store logs, enhancing trust in third-party management information systems, and has been implemented in the Portuguese judicial system. Its Hyperledger Fabric-based consensus ensures tamper-proof logging, but permissioned networks risk insider attacks if access controls fail.

In contrast, BlockBlend [10] proposed a solution to achieve timely, efficient, transparent and fair judicial delivery in the Indian judicial system through blockchain technology, focusing on addressing the issue of case backlog. Liu et al. [11] addressed the efficiency and security issues of blockchain consensus mechanisms in the judicial system by proposing an enhanced Bitcoin-NG consensus mechanism. This mechanism reduces the consensus time of blockchain networks, achieving an efficient judicial depository. These solutions either focus on the digital storage of court proceedings or propose blockchain-based solutions to other problems in the justice system, with few solutions focusing on the preservation and validation of electronic evidence for litigation.

III. Preliminaries

A. Elliptic Curve Cryptography

Building decentralized transactions on the blockchain requires an encryption algorithm where the transaction initiator digitally signs transaction data using their private key. Transaction validators only need to know the initiator's public key to verify the validity of the transaction, ensuring that it indeed originates from the initiator. This scenario in cryptography is known as *public key cryptography*, also known as *asymmetric cryptography* [12].

Common public key algorithms include RSA and elliptic curve cryptography (ECC). RSA [13], having an earlier inception, is widely used in the field of asymmetric cryptography, such as in the majority of SSL certificates. On the other hand, ECC algorithms [14], [15], due to their superior resistance to attacks and resource efficiency, are becoming increasingly prevalent. Asymmetric cryptographic algorithms are typically based on mathematical problems, with RSA relying on the difficulty of integer factorization and ECC based on the discrete logarithm problem on elliptic curves. Further theoretical insights can be found in [14].

ECC encompasses various cryptographic algorithms, including elliptic curve integrated encryption scheme (ECIES), elliptic curve Diffie–Hellman (ECDH), elliptic curve digital signature algorithm (ECDSA), and others [16]. Due to the utilization of different elliptic curves, elliptic curve algorithms adhere to multiple standards, such as:

1) NIST standards, with typical curves like P-256/384/521.
2) SECG standards, including typical curves such as Secp256k1/Secp256r1/Secp192k1/Secp192r1.
3) ECC25519, primarily referring to the Ed25519 digital signature and Curve25519 key agreement standards.
4) Cryptographic algorithm standards formulated by China's State Cryptography Administration, with typical examples such as SM2/3/4.

B. Secure Hash Algorithm

Secure hash algorithm (SHA) computes an irreversible unique data hash value through logical operations on file content data [17], [18]. In blockchain architecture, the ledger layer embeds the digest value of the previous block into the next block to form a blockchain-like data structure, ensuring the integrity and authenticity of the data. This is the basis for the tamper-proof and traceable characteristics of blockchain systems. Before storing data on the blockchain, blockchain systems can preserve sensitive data or files by storing their data fingerprints using message digest methods on the blockchain. Data can be protected by secure encryption methods to safeguard sensitive privacy data.

The principle of a hash function is to divide input data into blocks of equal size, padding if necessary, and then append a length block. Subsequently, based on the hash value of each block and the previous block, the hash value of the current block is calculated, with the output of the last block serving as the hash digest of the entire message. Commonly used hash algorithms include SHA-1, SHA-2, and SHA-3, among which SHA-2 is categorized into algorithms such as SHA-224, SHA-256, SHA-384, and SHA-512, based on the length of the message digest [19].

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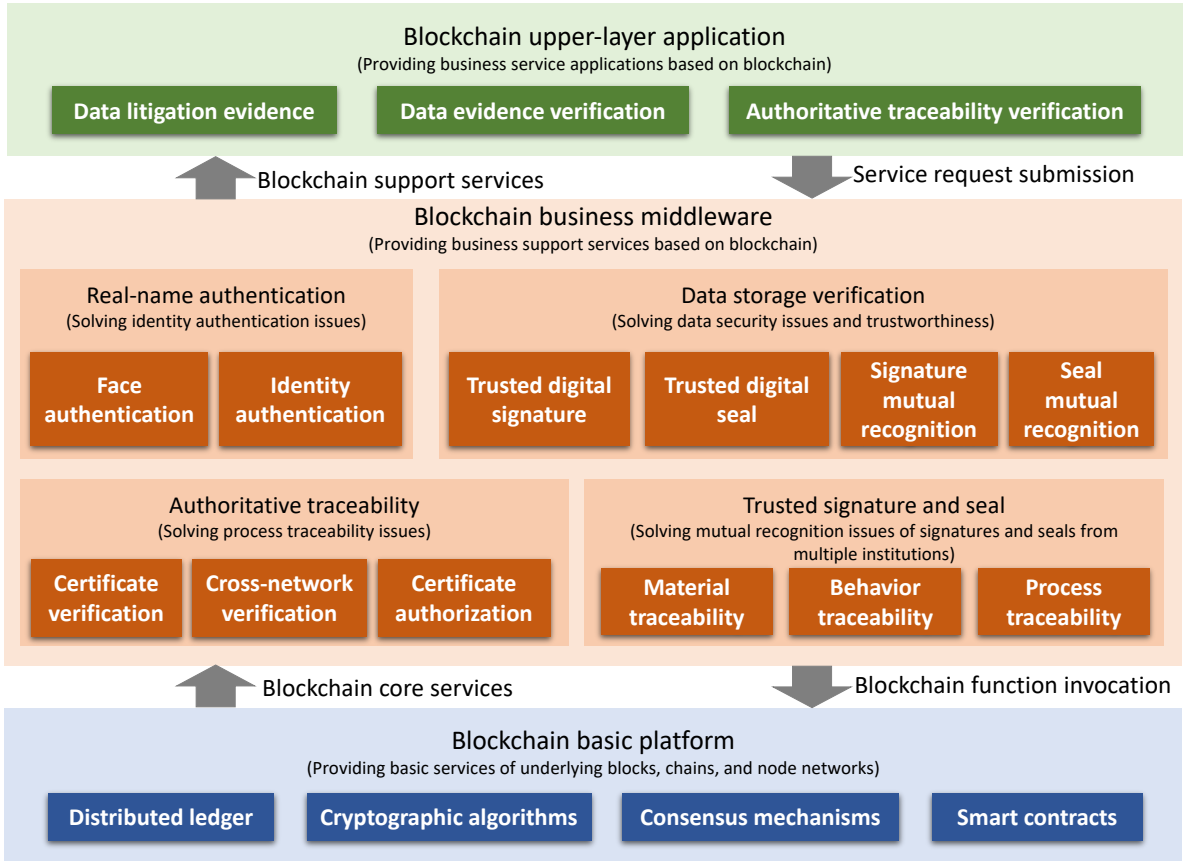


Fig. 1. System overview of SecEvidence.

IV. System Design

A. System Overview

Our system has the following characteristics:

- **Neutrality**. By forming a consortium blockchain with multiple institutions, utilizing the decentralized nature of blockchain along with a multi-party supervision mechanism, a tamper-proof database is established. Even courts cannot unilaterally tamper with data, ensuring the integrity and credibility of the data.
- **Security**. The blockchain network employs asymmetric cryptography to ensure the security of the system and the privacy of on-chain data, while also enabling the scrutiny of on-chain data content. Code replaces text, adhering to the principle of transparency, ensuring cross-chain trust and external link security, and safeguarding the authenticity of evidence rules.
- **Controllability**. Our system implements a permission authorization mechanism, establishes comprehensive access and management standards, as well as supervision, review, and exit mechanisms, ensuring the controllability and stability of the system.
- **Openness**. With adherence to access standards, the system opens its interface to society, supporting access to electronic data certification and other consortium blockchains that meet the standards, promoting openness and sharing of the system.

The system overview of SecEvidence, as depicted in Fig. 1, consists of three parts: the blockchain basic platform, the blockchain business middleware, and the blockchain upper-layer application scenarios. The details are as follows:

1) *Basic platform*. Comprises blockchain underlying technologies such as distributed ledger, cryptographic algorithms, consensus mechanisms, smart contracts, etc.
2) *Business middleware*. Deeply integrates litigation electronic evidence with blockchain technology, serving as the *bridge* for data interaction between court business systems and underlying platforms.
3) *Upper-layer applications*. Based on the Internet judiciary service alliance, providing scenarios such as internet electronic data litigation evidence preservation and evidence verification.

B. Concrete Construction

1) System architecture: The process of electronic evidence preservation and authentication services for SecEvidence is shown in Fig. 2. These services include litigation online certification, litigation process certification, and evidence authenticity certification, are built upon the blockchain business

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middleware described in Fig.1. Specifically, they leverage real-name authentication, data storage verification, authoritative traceability, and trusted signature and seal mechanisms to ensure the security, cross-institution recognition, and credibility of electronic evidence throughout the litigation lifecycle.

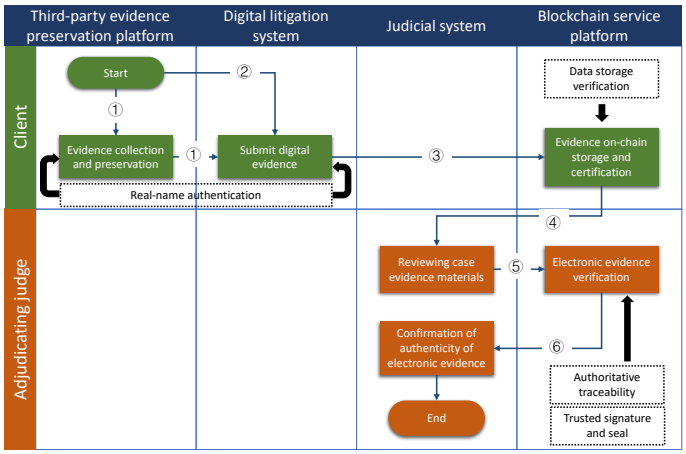


Fig. 2. Electronic evidence preservation and authentication services of SecEvidence.

Provide litigation online certification services. The general public can upload electronic evidence materials through affiliated third-party certification platforms, thereby solidifying the evidence. In the event of litigation, the submitted electronic evidence can be authenticated to demonstrate that it has remained unchanged since the time of certification.

Provide litigation process certification services. The system promptly certifies evidence materials submitted by parties to the court by linking them to the blockchain. If a party submits physical materials, they are synchronized to the blockchain when the court scans and generates electronic case files. Conversely, if the party submits electronic materials, the electronic data is directly linked to the blockchain for certification. When litigants or their legal representatives file a lawsuit application through the court's electronic litigation platform, they can simultaneously submit relevant electronic evidence materials. Such electronic evidence materials can be integrated into the blockchain platform. After the lawsuit application is approved, the certified electronic evidence materials can be submitted to the presiding judge as reliable evidence. If litigants submit paper materials directly to the court for case filing, after the case filing window approves the submission, the paper materials are transferred to the scanning center for case file scanning. During the scanning, categorization, and indexing of litigation materials, the litigation materials are solidified by linking them to the blockchain. Additionally, they are integrated into the electronic case management system. For litigation materials that have been linked to the blockchain, after creating the electronic case file, a blockchain linkage identifier is automatically added for display.

Provide evidence authenticity certification services. Litigants and judges can conduct online verification of the electronic case files stored on the certification platform, thereby confirming that the materials have not been tampered with since their initial creation. During the case proceedings, if the presiding judge or the opposing party doubts the authenticity of electronic evidence, they can utilize third-party certification platforms to authenticate the evidence's authenticity.

According to the Byzantine fault tolerance (BFT) algorithm [20], using $3f + 1$ nodes can tolerate up to $1/3$ Byzantine faulty nodes. Therefore, blockchain construction requires at least 4 nodes for achieving consensus. Here, f represents faulty Byzantine nodes. This implies that in the worst-case scenario, only one node failure is allowed. In this situation, even if one node experiences issues (no feedback, feedback of incorrect data, or unstable feedback data), the other three nodes can still achieve consensus, ensuring the system's normal operation and data consistency.

2) Algorithm design: **Elliptic curve algorithm**. The elliptic curve algorithm design in this paper and project employs the ECDSA signature algorithm, which offers high security performance and maintains cryptographic efficiency. ECDSA first requires establishing a finite field $\mathbb{F}_p$, where p is a prime number, abstracting the elliptic curve on this finite field, where all points on the finite field are discrete points. Basic operations and elements such as addition, inverse operations, and zero elements are defined for all discrete points on the elliptic curve in this finite field.

In this paper, SM2P256V1 is employed instead of Ed25519 to align with specific application scenarios and security standards, represented as [21]:

$$y^2 = x^3 + ax + b. \tag{1}$$

where a = FFFFFFFE FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF 00000000 FFFFFFFF FFFFFFFC, and b = 28E9FA9E 9D9F5E34 4D5A9E4B CF6509A7 F39789F5 15AB8F92 DDBCBD41 4D940E93. For prime numbers we choose p = FFFFFFFE FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF 00000000 FFFFFFFF FFFFFFFF FFFFFFFF.

Initially, define the addition of points on the elliptic curve within the real number range. Adding two points involves drawing a line connecting them and intersecting this extended line with the curve. The point of intersection mirrored along the x-axis on the curve is the result of adding the two points. Based on the addition definition in the real number range, we can derive the addition formula for known points P and Q:

$$\Delta = \frac{Y_Q - Y_P}{X_Q - X_P} \tag{2}$$

$$X_R = \Delta^2 - X_P - X_Q, \tag{3}$$

$$Y_R = -Y_P + \Delta(X_P - X_R), \tag{4}$$

When P and Q are the same point being added, we have:

$$\Delta = (3X_P^2 + a)/(2Y_P). \tag{5}$$

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In a finite field $\mathbb{F}_p$, where p is a prime number, the corresponding addition and multiplication operations become addition and multiplication operations in the finite field. The addition calculation formula is as follows [22]:

$$\lambda = \frac{Y_Q - Y_P}{X_Q - X_P} \bmod p, \tag{6}$$

$$X_R = \lambda^2 - X_P - X_Q \bmod p, \tag{7}$$

$$Y_R = -Y_P + \lambda(X_P - X_R) \bmod p, \tag{8}$$

When adding the same point, we have:

$$\lambda = (3X_P^2 + a)/(2Y_P) \bmod p. \tag{9}$$

The basic parameters of the curve, including the base point G (which can generate a cyclic group of the elliptic curve over the finite field) and the order of G, are publicly known.

For nodes, the private key is a number d less than the order of the curve, and the public key is generated based on the private key, i.e., dG. According to the discrete logarithm principle, dG can be easily computed from d. However, for points on an elliptic curve, given dG and G, it is not possible to directly calculate d. This implies that the private key's security is preserved because it cannot be derived from the public key.

Secure hash algorithm. This paper adopts the SM3 hash function [23] as an alternative to the original hash function to better satisfy specific security and compliance requirements. The computational structure of SM3 is similar to that of SHA-256: it first pads the input, and then iteratively applies a compression function. In each iteration, the compression function takes the previous round's result and the current message block as inputs, performs an *XOR* operation with the previous result, and produces the output for the current round. The final iteration output serves as the output of the hash algorithm.

V. Experimental Evaluation

A. Experimental Setup

This experiment was conducted on a platform with 2GB of memory, Intel(R) Xeon(R) Platinum 8269CY CPU @ 2.50GHz, running Ubuntu 18.04 LTS. The development was done using the Golang language, with tendermint v0.33.2 as the core module. It consists of two main modules: tendermint core, which includes the primary blockchain consensus mechanism, and tendermint ABCI (Application Blockchain Interface), which provides an interface for developing blockchain applications based on tendermint [24]. GMSSL 2.5.4 and OpenSSL 1.1.0d were used as underlying libraries for security algorithms.

B. Performance Comparisons

The test data are shown in TABLE I, we conducted 10 rounds of inserting a transaction and completing the block creation process for different algorithms. We record the time taken for all signature, verification, and hash digest calculations. The last test case was used to assess the number of calls to each algorithm under repeated input conditions.

TABLE I
Test data of SecEvidence.

ID	Test Input
1	name=aaa
2	name=bbb
3	name=ccc
4	name=ddd
5	name=eee
6	name=fff
7	name=ggg
8	name=hhh
9	name=iii
10	name=iii

Our results are presented in TABLE II and TABLE III, demonstrating that we can obtain the same output as the default secure algorithm. After upgrading the algorithm security, the system's invocation time for the corresponding algorithm differs from the default secure algorithm by no more than 2 milliseconds, meeting the basic requirements of permissioned chains. These time costs are primarily influenced by algorithm performance rather than changes to the algorithm interface, and there is no significant delay in the block generation process. In comparison, network transmission latency costs are higher. Test results under erroneous input conditions show that the system only needs to perform one hash digest algorithm operation, without involving signature and block generation operations. Under error conditions, the time cost of hash digest is approximately twice the average time.

TABLE II
Comparison of average and maximum time at legitimate input for each algorithm.

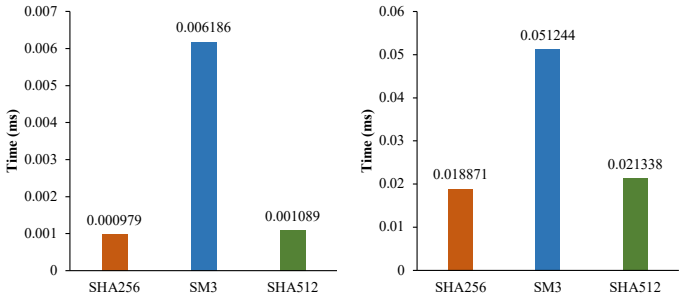
Algorithms	Average Time (ms)	Maximum Time (ms)	Number of Executions
Ed25519-Sign	0.055129	0.066625	6
Ed25519-Verify	0.149707	0.214072	14
SM2P256V1-Sign	1.247249	1.310879	6
SM2P256V1-Verify	0.051597	0.087164	14
SHA256	0.000979	0.018871	1120
SM3	0.006186	0.051244	1120
SHA512	0.001089	0.021338	1120

TABLE III
Comparison of time to repeat input for each algorithm.

Algorithms	Average Time (ms)	Number of Executions
Ed25519-Sign	-	0
Ed25519-Verify	-	0
SM2P256V1-Sign	-	0
SM2P256V1-Verify	-	0
SHA256	0.001965	1
SM3	0.002450	1
SHA512	0.018479	1

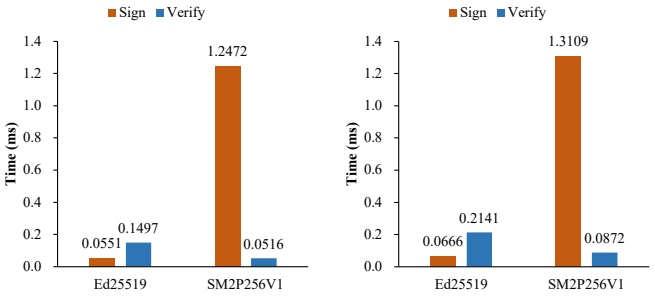
The comparison of average and maximum time for hash algorithms is shown in Fig. 3. The results indicate that the time cost of the high-security SHA512 and SM3 algorithms is

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(a) Average time (b) Maximum time

Fig. 3. Comparison of average and maximum time for hash algorithms, where (a) denotes the average time, (b) denotes the maximum time.



(a) Average time (b) Maximum time

Fig. 4. Comparison of average and maximum time for asymmetric cryptography algorithms. (a) denotes the average time, (b) denotes the maximum time.

higher than that of the lower-security SHA256 algorithm, which is in line with expected estimates. The comparison of time overhead for elliptic curve signature algorithms is illustrated in Fig. 4. The time overhead of the SM2 signature algorithm is significantly higher than that of the Ed25519 algorithm, while the time overhead of the verification algorithm is lower than that of the Ed25519 algorithm. Therefore, users can choose different algorithms based on the efficiency of signature and verification calls during the ABCI implementation process.

VI. Conclusion

In conclusion, `SecEvidence` presents a robust solution to ensure the security and credibility of electronic evidence in litigation scenarios. By leveraging blockchain technology and cryptographic techniques, it addresses the challenges posed by traditional electronic data vulnerabilities. Our experiments demonstrate that enhancing critical algorithm security does not compromise system performance. `SecEvidence`'s design, incorporating multi-node consensus and cryptographic methods, ensures tamper-proof, verifiability, and traceability of electronic data evidence. We believe that `SecEvidence` contributes significantly to enhancing the security and credibility of electronic litigation services, thereby fostering trust and confidence in the judicial process.

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References

[1] A. Gupta, "E-commerce: Role of e-commerce in today's business," *International Journal of Computing and Corporate Research*, vol. 4, no. 1, pp. 1–8, 2014.

[2] V. Santos, T. Augusto, J. Vieira, L. Bacalhau, B. M. Sousa, and D. Pontes, "E-commerce: issues, opportunities, challenges, and trends," *Promoting organizational performance through 5G and agile marketing*, pp. 224–244, 2023.

[3] X. Zhang *et al.*, "The dilemma and countermeasures of the judicial application of electronic contracts in the digital age," *Academic Journal of Biochemistry and Molecular Biology*, vol. 1, no. 2, pp. 1–9, 2022.

[4] Z. Zheng, S. Xie, H. Dai, X. Chen, and H. Wang, "An overview of blockchain technology: Architecture, consensus, and future trends," in *IEEE international congress on big data*, 2017, pp. 557–564.

[5] M. Stevens, E. Bursztein, P. Karpman, A. Albertini, and Y. Markov, "The first collision for full sha-1," in *Annual International Cryptology Conference*, 2017, pp. 570–596.

[6] S. Haridas, S. Saroj, S. T. Maddala, and M. Kiruthika, "Legalledger–blockchain in judicial system," in *Innovative Data Communication Technologies and Application*, 2022, pp. 471–482.

[7] A. Verma, P. Bhattacharya, D. Saraswat, and S. Tanwar, "Nyaya: Blockchain-based electronic law record management scheme for judicial investigations," *Journal of Information Security and Applications*, vol. 63, p. 103025, 2021.

[8] A. Ranjan, A. N. Singh, A. Kumar, B. Prashanth, and M. M. Kumar, "Transforming judicial systems with blockchain: A court case governance system for tamper-proof and transparent legal processes," in *International Conference on Applied Intelligence and Sustainable Computing (ICAISC)*, 2023, pp. 1–7.

[9] R. Belchior, M. Correia, and A. Vasconcelos, "Justicechain: Using blockchain to protect justice logs," in *On the Move to Meaningful Internet System*, 2019, pp. 318–325.

[10] N. Khairwal and D. Shah, "Blockblend: An interoperable decentralized framework for court excellence," in *International Conference on Advances in Science and Technology (ICAST)*, 2022, pp. 16–21.

[11] S. Liu and Q. Zheng, "Research on blockchain consensus mechanism for judicial depository applications," in *International Conference on Electronic Information Engineering and Computer Science (EIECS)*, 2023, pp. 913–918.

[12] W. Diffie and M. E. Hellman, "New directions in cryptography," in *Democratizing Cryptography: The Work of Whitfield Diffie and Martin Hellman*, 2022, pp. 365–390.

[13] R. L. Rivest, A. Shamir, and L. Adleman, "A method for obtaining digital signatures and public-key cryptosystems," *Communications of the ACM*, vol. 21, no. 2, pp. 120–126, 1978.

[14] N. Koblitz, "Elliptic curve cryptosystems," *Mathematics of computation*, vol. 48, no. 177, pp. 203–209, 1987.

[15] V. S. Miller, "Use of elliptic curves in cryptography," in *Conference on the theory and application of cryptographic techniques*, 1985, pp. 417–426.

[16] N. Saqib and U. Iqbal, "Security in wireless sensor networks using ecc," in *IEEE International Conference on Advances in Computer Applications (ICACA)*, 2016, pp. 270–274.

[17] D. Eastlake 3rd and P. Jones, "Us secure hash algorithm 1 (sha1)," Tech. Rep., 2001.

[18] W. Penard and T. van Werkhoven, "On the secure hash algorithm family," *Cryptography in context*, pp. 1–18, 2008.

[19] S. Debnath, A. Chattopadhyay, and S. Dutta, "Brief review on journey of secured hash algorithms," in *International Conference on Opto-Electronics and Applied Optics (Optronix)*, 2017, pp. 1–5.

[20] L. Lamport, R. Shostak, and M. Pease, "The byzantine generals problem," in *Concurrency: the works of leslie lamport*, 2019, pp. 203–226.

[21] C. S. T. C. of China, "GM/T 0003.2-2012: Public Key Cryptograhpic Algorithm SM2 Based on Elliptic Curves - Part 5: Parameter Definition," 2012, in Chinese.

[22] W. Mao, *Modern cryptography: theory and practice*. Pearson Education India, 2003.

[23] ISO/IEC, "IT Security techniques Hash-functions – Part 3: Dedicated hash-functions," International Organization for Standardization, Tech. Rep. ISO/IEC 10118-3:2018, 2018, Available at https://www.iso.org/standard/67116.html.

[24] Tendermint. (2024) Tendermint documentation. [Online]. Available: https://docs.tendermint.com/v0.33/

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